

Report

For this project, I worked with the Breast Cancer Wisconsin dataset to predict whether a tumor is malignant or benign based on 30 numeric features radius, texture, area etc.

I started by importing libraries and dataset. After cleaning the data, I mapped malignant = 1 & benign = 0. I split dataset into training 80 % and testing 20% , so the models could learn from one part of the data and be tested on data they'd never seen before.

After that I trained five different models:

1. Logistic Regression
2. Decision Tree
3. K-Nearest Neighbors
4. Support Vector Machine
5. Naive Bayes

Each model was evaluated using accuracy, precision, recall and F1 score. Since this is a medical diagnosis problem so recall was given the most importance because missing an actual cancer case is far more dangerous than marking a healthy patient as to be sick. The Naive Bayes model performed the best with highest accuracy of 97.4%.

After that I did cross validation on the Logistic Regression model which gave an average accuracy of 95.4%. Decision Tree showed a big gap between train 100% and test 94.7% accuracy meaning it had over-fit. By GridSearchCV its max_depth came out best at 3. After training it again , the train accuracy dropped to 97.8% while test accuracy remained the same so the gap reduced.

Overall, this project showed that different models perform differently and comparing them properly and checking them is important before selecting one for a real task.